FINANCE TRANSFORMATION · DESIGN AND WORKING PROTOTYPE

Agentic finance: from siloed processes to one governed flow

Order to cash, procure to pay, record to report, treasury and FP&A, redesigned so AI agents propose and people decide

Shubham Khadria · October 2026

EXECUTIVE SUMMARY

Agents can run the routine of finance; people should own judgement, approvals and the outliers

The problem	Finance runs as separate processes joined by spreadsheets, email and month-end catch-up. AI adoption is high, but few teams have scaled it with results they trust.
The design	One pattern across five processes: rules and the ledger stay authoritative, agents work the intake and the exceptions, people approve and decide.
The proof	A working cash application prototype on synthetic data: 93% of payments applied automatically at 100% accuracy, every posting logged with who approved it.
The path	Domain by domain: fix workflow debt first, prove accuracy, then widen what agents may do on their own as trust is earned.

WHY NOW

Finance has adopted AI quickly, but few teams have scaled it with results they trust

44%

of CFOs used gen AI in more than five use cases in 2025, up from 7% a year earlier

McKinsey, Nov 2025

15–25%

of CFOs have scaled AI across their finance functions

Bain CFO survey, Apr 2026

86%

of finance teams have run into inaccurate or made-up AI output

Maximor survey of 100 CFOs, in Journal of Accountancy, Feb 2026

40%+

of agentic AI projects will be cancelled by the end of 2027

Gartner, cited by Deloitte, Mar 2026

The gap is scaling with trust, not adoption. This design is built around that gap: accuracy before automation, and every action auditable.

DESIGN PRINCIPLES

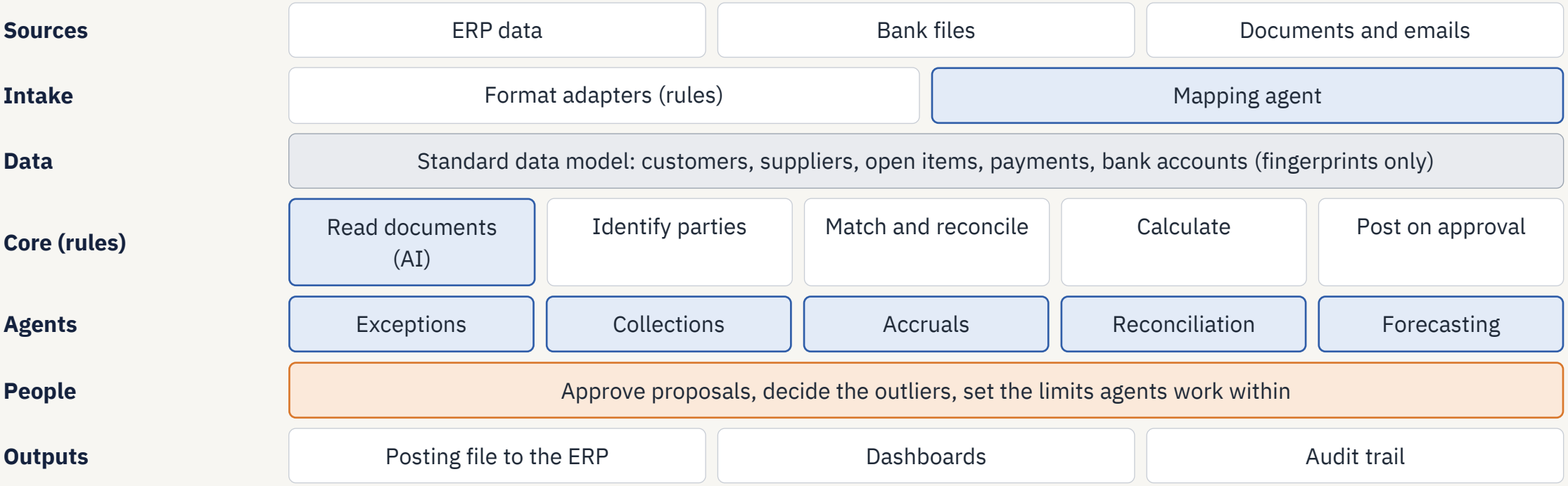
Five design principles, consistent with current thinking from McKinsey, Bain and others

Principle	What it means in this design	Where the idea appears
1 · Reinvent workflows end to end	Redesign each process from intake to posting, not one step at a time	McKinsey: process reinvention, vertical use cases
2 · Pay down workflow debt first	Standard data model, format adapters and clean master data before agents	Bain: pay down workflow debt before deploying agents
3 · Governed autonomy	Agents propose; rules and the ledger stay authoritative; people set the limits	McKinsey: governed autonomy in the agentic AI mesh
4 · In the loop, then above it	People approve each action now, then supervise outcomes as accuracy is proven	McKinsey: human in the loop vs above the loop
5 · Auditable by design	Every proposal, approval and agent action is logged with its evidence	Journal of Accountancy: CFOs trust AI they can audit

TARGET ARCHITECTURE

One architecture: AI at the edges where the variety is, rules at the core, people on the outliers

Common to all five processes; slide 7 shows it in cash application



Blue border: AI agent · Orange border: people decide · Grey border: rules and data · 05

AUTONOMY LADDER

Autonomy is earned step by step: agents start by suggesting and act alone only within agreed limits

1 · Extract

Read documents into structured data

Remittances, invoices, contracts

2 · Suggest

Propose a match, a mapping or a fix

Cash matches, account mappings

3 · Investigate

Gather evidence and explain the case

Blocked invoices, revenue items

4 · Act with approval

Execute once a person approves

Credit memos, payment runs

5 · Act within limits

Execute alone inside agreed limits

Exact matches, small write-offs

Human in the loop: a person approves each action

Above the loop: people supervise

ORDER TO CASH · BUILT

Cash application that posts only what a person or an agreed rule approves



93–98%
of payments applied automatically across test runs

100%
accuracy of automatic matches against the answer key

0
payments put on the wrong customer

$0.00
gap between cash applied and the drop in AR

Accuracy over automation. Uncertain matches go to review; a wrong posting costs more than a manual one.

Residual items. A short pay clears the invoice and opens a new item that keeps the original due date.

Protected bank data. Accounts stored as keyed-hash fingerprints plus last 4 digits only.

Prototype on synthetic data: 46 bank payments, 1,000 invoices · Python, SQL, Streamlit, open LLMs via Ollama · 07

ORDER TO CASH · DESIGNED

Three agents work the outliers in order to cash and hand off to each other

Deductions triage

Starts when a short pay is posted as a residual item

Does on its own

Classifies the reason, fetches backup for that claim only, drafts the credit or the denial

A person decides

Credits and write-offs, with the approver set by amount

Invalid claims go to collections

Remittance chase

Starts when cash arrives without a remittance advice

Does on its own

Emails the payer on day 0, reminds midway, passes replies to the remittance AI

A person decides

Anything still unresolved on day 5

Unresolved items go to the AR team

Collections

Starts every morning, across all open items

Does on its own

Ranks customers on past dues, risk and DSO impact; sends template reminders to low-risk accounts

A person decides

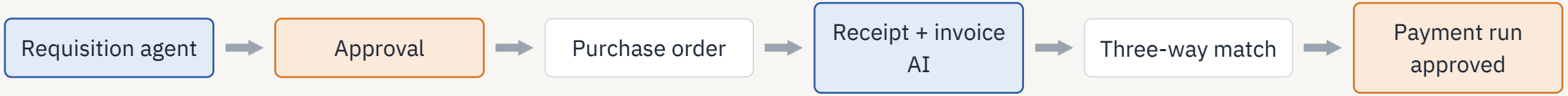
Collectors call the priority list, each with a call brief

New disputes go to deductions

Amounts under valid dispute are excluded from collections, so collectors chase only what is collectible · 08

PROCURE TO PAY · DESIGNED

Procure to pay mirrors order to cash: rules match order, receipt and invoice; agents chase the gaps



Order to cash (built)	Procure to pay (mirror)
Bank file and remittance	Supplier invoice
Remittance AI	Invoice AI
Payer ID by bank account	Supplier ID and bank-detail check
Cash-to-invoice matcher	Three-way match
Residual item for a short pay	Invoice blocked for a variance
Collections agent	Payment agent (discounts)

Exceptions agent: a price gap goes to the buyer; a quantity gap asks the receiver to confirm delivery.

Top cause of stuck invoices: no receipt for services, so the agent nudges service owners.

Top fraud risk: fake supplier bank-detail changes, always verified independently.

RECORD TO REPORT · DESIGNED

A continuous close finds breaks every day, so the period end keeps only what needs a closed period

Every day during the month

- Reconciliation to the GL
- Service accruals
- Intercompany matching
- Journal entry review
- Change watcher for revenue
- Revenue exceptions
- Account mapping
- Close orchestrator
- **Daily close health report:** every account or item that does not agree with its source
- Accountants fix breaks the same day

At period end

- FX revaluation
- Allocations and depreciation
- Revenue recognition (RAR)
- Group reporting and consolidation
- Flux and commentary agent
- Controller signs off

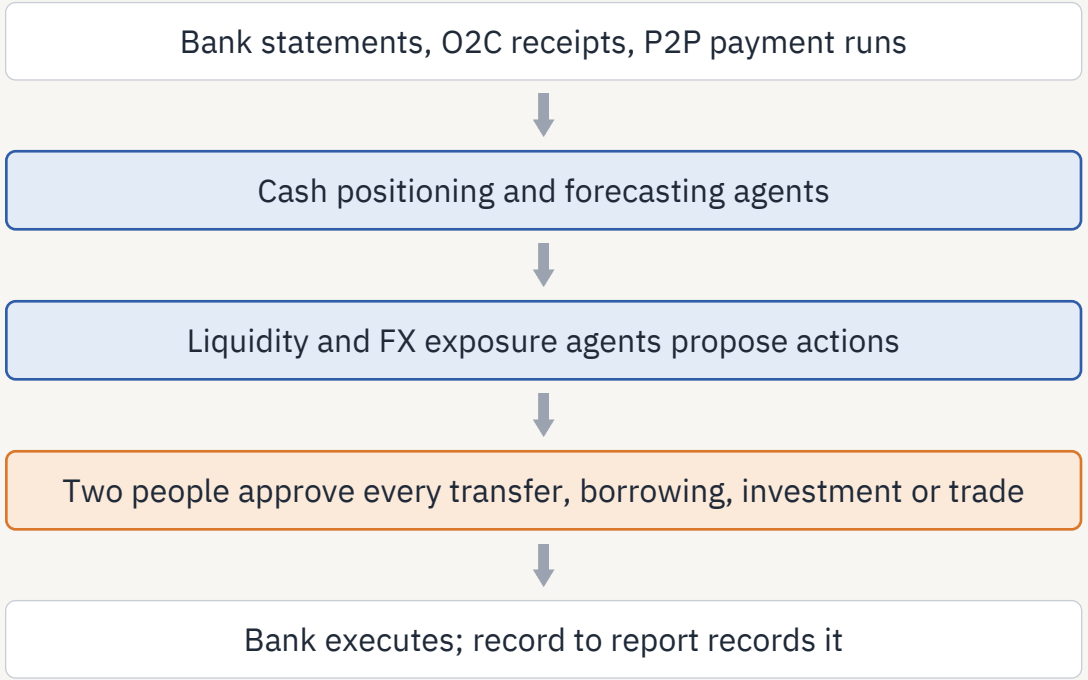
Usual hold-ups: chasing people, service accruals, upstream revenue changes, unmapped accounts · 10

TREASURY · DESIGNED

Order to cash and procure to pay feed one daily view of cash; agents never move money

Horizon	Question	View
Today	Where is our cash, and must we move it?	Cash position
13 weeks	Will we have enough each week, and where are the dips?	Cash forecast
12+ months	How do we fund the business and invest surplus cash?	Liquidity plan

The hardest input is already built: when customers will actually pay, predicted from their order-to-cash payment history.



FP&A · DESIGNED

FP&A: agents collect and reconcile the numbers so analysts can advise the business

- **Input collection:** pre-filled templates in one shared system; validates entries and chases late inputs
- **Driver model and scenarios:** the plan built from drivers; answers what-if questions in plain language
- **Rolling forecast:** updates continuously from actuals, sales pipeline, hires and the cash forecast
- **Variance explanation:** splits price, volume, mix and timing; flags accounting errors; drafts commentary
- **Allocations and cross-charges:** applies the agreed rules and checks both sides agree

Bain's three ways to modernize planning

Streamline
One plan, one version; inputs validated against history

Enhance
Variance explanations and plain-language what-ifs

Reinvent
Rolling, event-triggered planning instead of a fixed annual cycle

Analysts own assumptions and advise leaders

REUSABLE BUILDING BLOCKS

Four reusable building blocks carry the design across all five processes

Mapping agent

Learns a new layout or code and proposes the mapping; a person approves it once

Used for bank file formats, new GL accounts in group reporting, new products in revenue

Exceptions agent

Investigates what the rules could not settle and proposes a fix with its evidence

Used for unmatched cash, blocked supplier invoices, stuck revenue items, reconciliation breaks

Daily health checks

Compares every balance or item with its source, every day, not at month-end

Used for subledgers vs GL, GR/IR, intercompany, forecast vs actual cash

Decision log

Records every proposal, approval and agent action, by whom and when

Used everywhere: the basis for audit and for widening what agents may do alone

ROADMAP

Start where the data is ready, prove accuracy, then widen autonomy domain by domain

Phase 1 · Prove

Cash application and its exceptions in one entity; standard data model; decision log

Measure

Auto-apply rate, accuracy, unapplied cash, DSO

Phase 2 · Extend

Deductions and collections agents; P2P three-way match exceptions; mapping agent for new formats

Measure

Deduction cycle time, blocked invoices, cost per invoice

Phase 3 · Connect

Continuous close, cash forecasting and rolling forecast on the same data and log

Measure

Days to close, forecast accuracy, analyst time spent advising

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Traps to avoid, from McKinsey's finance research

Waiting for perfect data · transforming the whole function at once · pilots without a business-linked roadmap · neglecting change management · automating fragmented processes before simplifying them

SOURCES

Sources for the figures and terms used in this deck

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- McKinsey Podcast, “AI is everywhere. The agentic organization isn’t—yet,” Apr 2026
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- Deloitte, “CFO Guide to Tech Trends 2026,” Mar 2026 (includes the Gartner 40% cancellation forecast)
- Gartner press release on AI and CFO margin growth, Apr 2026
- Journal of Accountancy, “Agentic AI is handling more finance work — but can CFOs trust it?”, Feb 2026

Prototype results come from the author’s own build on synthetic data with an answer key; they are not client results.

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